MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
4400 TAILINGS MANAGEMENT FACILITY (TMF) SLOPE STABILITY ANALYSIS FOR LOM TSF EMBANKMENT

Document Number **MD-216-4400-EG-CV-CAL-1583**
Revision: **D01**
Revision Date: **31.10.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	ST03	CED01	44001	OU100684620	4
L&T Document Number:	O19045	450	C	K	24279	001	4

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	04.06.2020	FIRST ISSUE	ATCW	SNM	SJM	
A02	08.08.2020	Revised as per Employer's comments dated on 27.06.2020	ATCW	SNM	SJM	
A03	14.10.2020	Revised as per Employer's comments dated on 02.09.2020	ATCW	SNM	SJM	
C01	01.12.2020	Issued for Construction	ATCW	SNM	SJM	
D01	31.10.2023	AS BUILT	ATCW	SNM	SJM	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute
TMF	:	Tailings Management Facility

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Slope Stability Analysis for LOM TSF Embankment.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Slope Stability Analysis for LOM TSF Embankment only.

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ATTACHMENT 1 SLOPE STABILITY ANALYSIS FOR LOM TSF EMBANKMENT

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ATTACHMENT 1.1 RADIUS OF TSF EMBANKMENT 500 M

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Job No.:	117168.03			
Calcs:	Bhagaban AC	Date:	Sep-20	
Checked:	KDS	Date:	Sep-20	

Project: Mansourah Massarrah Gold Project

Subject: Slope Stability Analyses for Start-Up TSF Embankment (Radius of Start-UP TSF Embankment = 500 m)

Software: Slope/W (GeoStudio)

Slope Stability Analyses of Startup TSF Facility

Height of Startup Embankment = 6 m

(Material parameters are assumed conservatively based on the ATCW database for the similar materials)

Layers	Unit Weight	Static / Post OBE		Post SEE	
	γ (kN/m^3)	ϕ (Deg)	s_u/σ'_v	ϕ (Deg)	s_u/σ'_v
1 Foundation (Gravelly Sand)	19	34	-	28.3	0.09
2 Bedding Layer (Sand)	16	23	-	18.8	0.09
3 Embankment (Rock)	20	35	-	29.3	-
4 Embankment (Sand)	18	27	-	22.2	-
5 Liner	10	10	-	8	-
6 Tailings	16	-	0.2		0.08

Stability Analyses Criteria:

Analyses Scenario	Factor of Safety Requirement (ANCOLD)
1 Static (Long-term - normal operating conditions)	FoS ≥ 1.5
2 OBE (Serviceability)	FoS ≥ 1.3
3 SEE (Safety)	FoS ≥ 1.2

Earthquake Acceleration (Ref: Seismic Hazard Assessment - MD-2160000-EG-MG-RPT-0007)

1 OBE Earthquake	PGA = 0.078 g
2 SEE Earthquake	PGA = 0.13 g

Result Output:

A: Embankment Stability Analyses (Ht of Embankment = 6.0 m)

Analyses Scenario	Downstream Failure FoS	Figure	Global FoS	Figure
1 Static Case (End of construction)	2.34	B1.1	4.16	B1.2
2 OBE Case	1.81	B2.1	2.84	B2.2
3 SEE Case	1.48	B3.1	2.40	B3.2

B: Ramp Stability Analyses (Ht of Ramp = 9.0 m)

Analyses Scenario	Downstream Failure FoS	Figure	Global FoS	Figure
1 Static Case (End of construction)	1.53	B4.1	2.15	B4.2
2 OBE Case	1.33	B5.1	1.70	B5.2
3 SEE Case	1.09	B6.1	1.41	B6.2
4 Static Case (with tailings)	4.46	B7.1	9.62	B7.2
5 OBE Case (with tailings)	2.16	B8.1	2.72	B8.2
6 SEE Case (with tailings)	1.35	B9.1	1.90	B9.2

C: TSF Stability Analyses with Tailings inside Embankment during Start-up (HOB = 9 m)

Analyses Scenario	Beach Slope Failure FoS	Figure
1 Static Case - Tailings Beach Slope Failure	12.61	B10
2 Post-OBE Case - Tailings Beach Slope Failure	10.08	B11
3 Post-SEE Case - Tailings Beach Slope Failure	4.36	B12

Note:

(1) Slope stability analyses result considering shallow (local) failure is not included here as it would have not significant influence in the overall stability as well as intrigity of the TSF embankment.

(2) Potential global failure as well as failure towards the downstream region of the TSF embankment is considered in the current stability analyses.

(3) Friction angle for the sand bedding layer is the assumed interface friction angle as this layer is underlain by the liner.

(4) Liner material has usually low friction angle thus 10 degree was assumed.

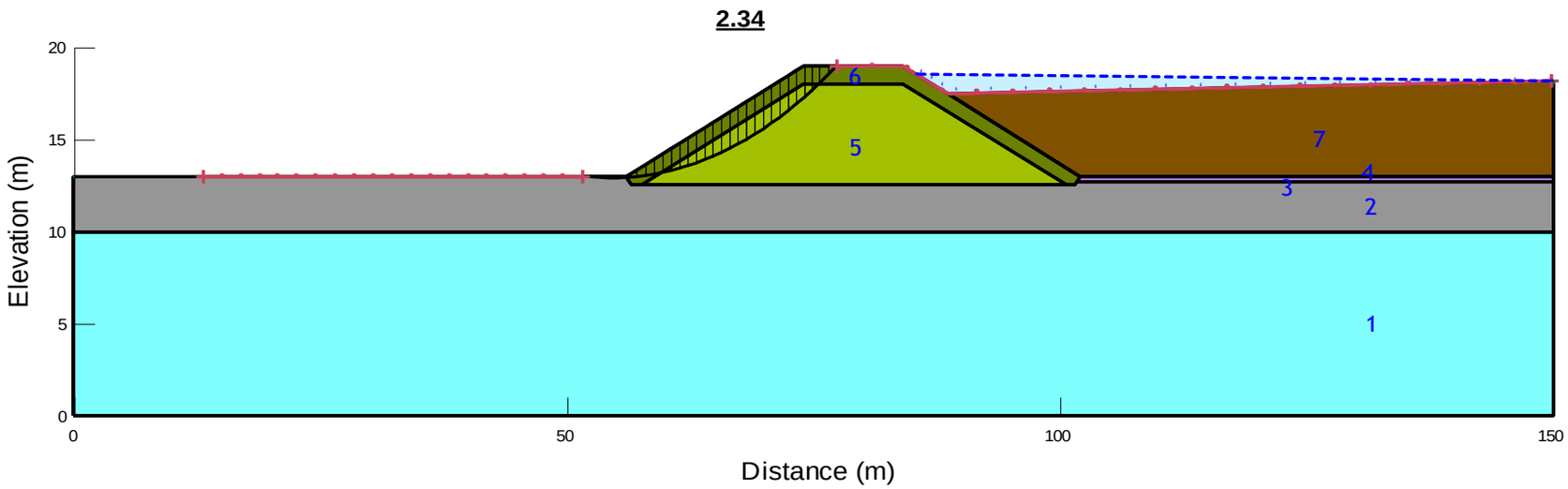
(5) For rockfill and embankment sand, friction angles were assumed conservatively based on the ATCW experiences in the similar materials.

(6) As per USACE approach, SEE analyses for embankment and ramp include reduction of strength parameters of materials by 20%, adopting liquefied strength of tailings, and adopting half of peak ground acceleration.

(7) For the TSF including tailings inside the embankments, post seismic analyses are conducted.

End

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Emb 6.0m_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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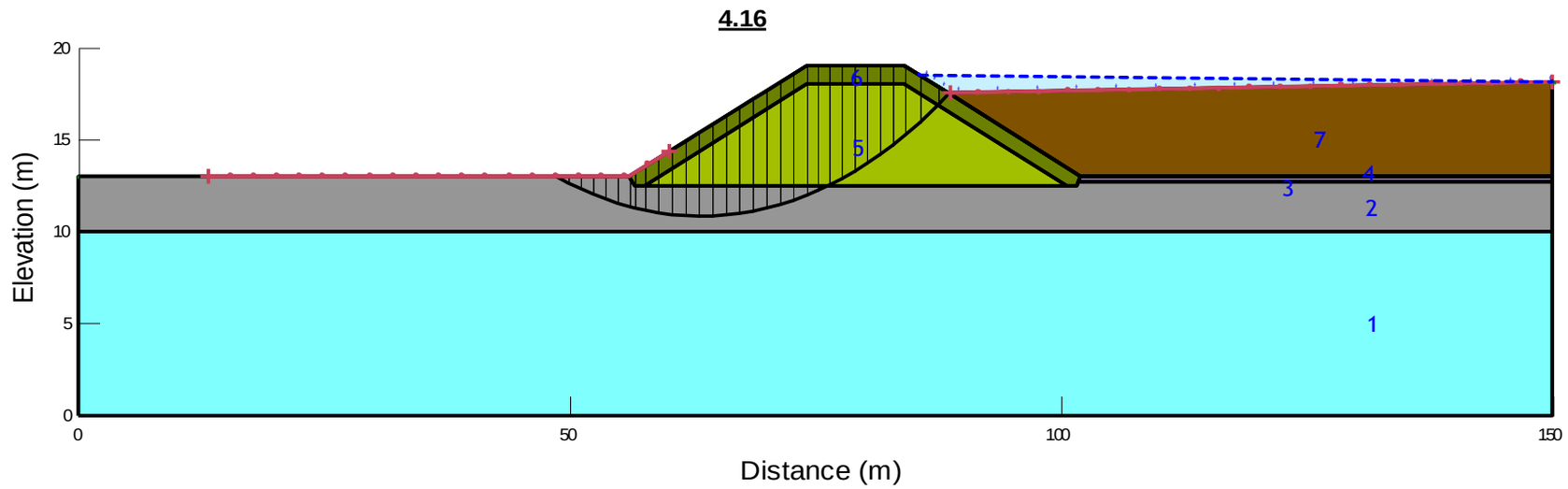
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Interior Embankment (Height 6.0 m)

Static Analyses (Embankment Downstream Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B1.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
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Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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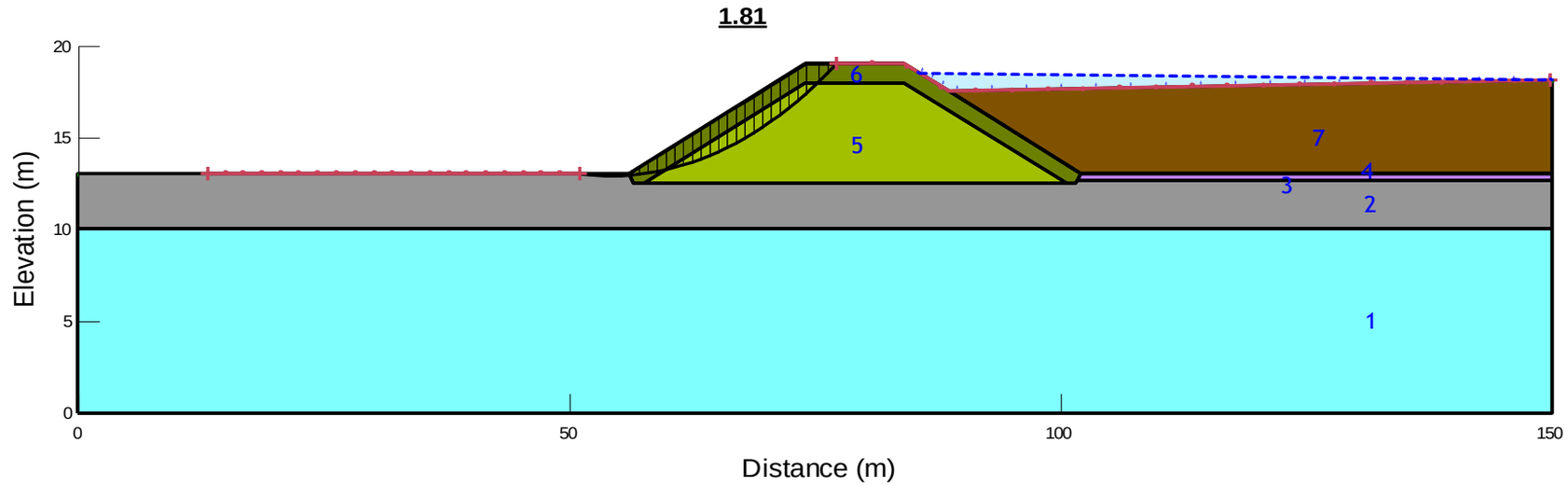
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Stability Analyses - Interior Embankment (Height 6.0 m)

Static Analyses (Global Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B1.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Emb 6.0m_OBE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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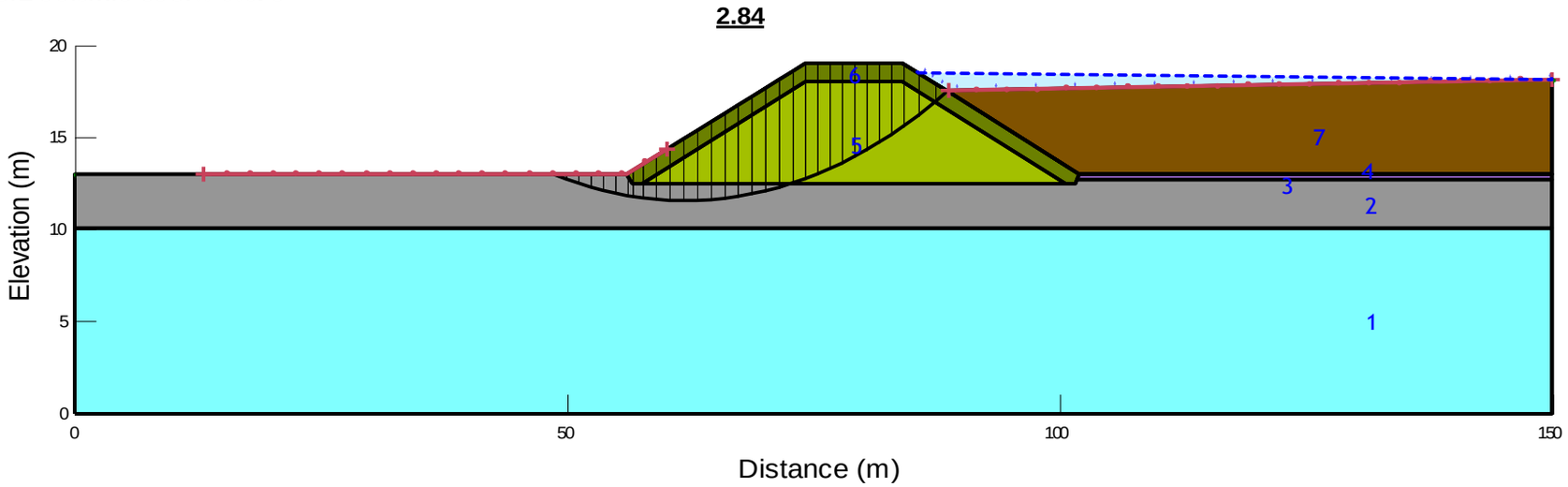
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Interior Embankment (Height 6.0 m)

OBE Analyses (Embankment Downstream Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B2.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
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Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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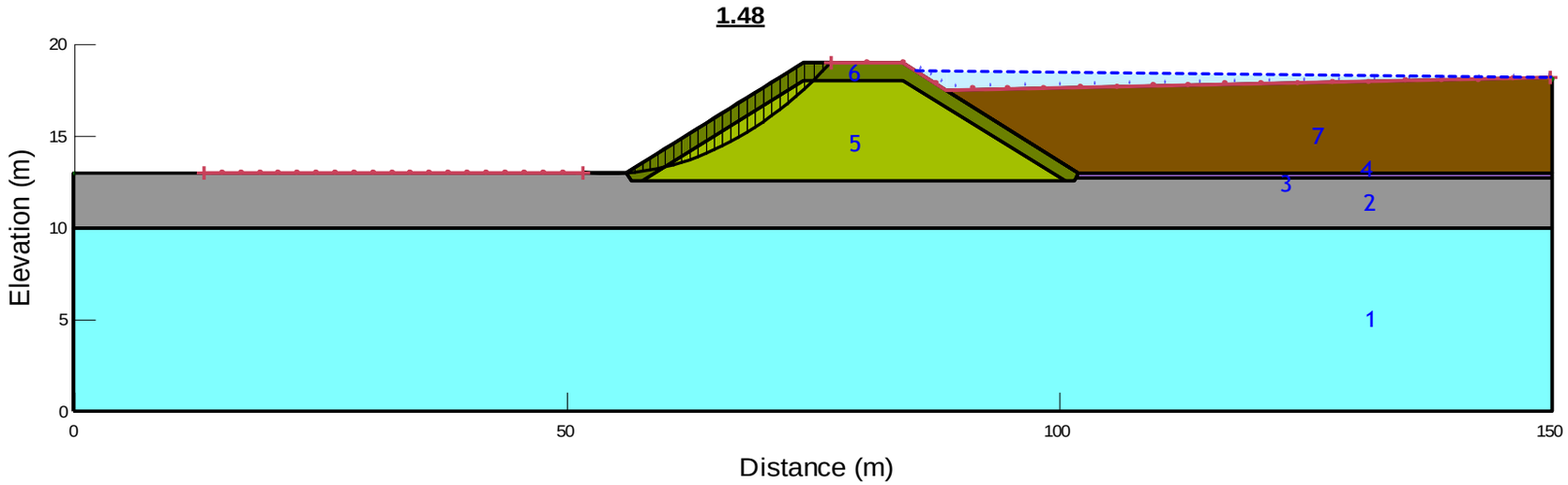
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Stability Analyses - Interior Embankment (Height 6.0 m)

OBE Analyses (Global Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B2.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Emb 6.0m_SEE.gsz
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Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 18.8 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Minimum Strength: 0
Piezometric Line: 1

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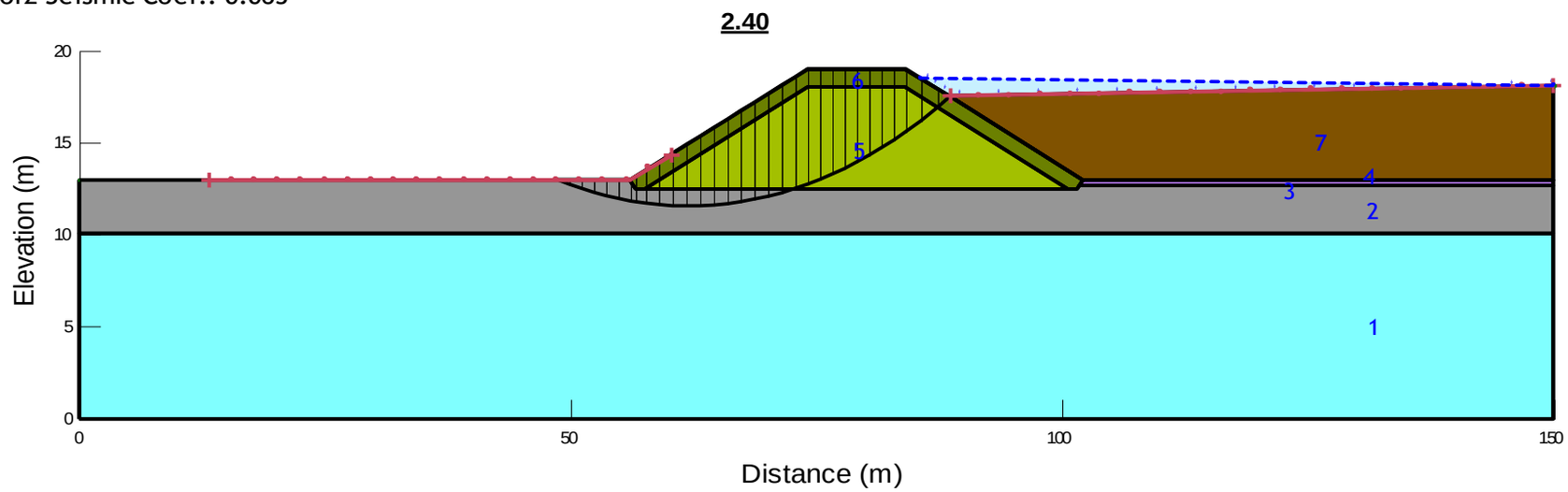
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Stability Analyses - Interior Embankment (Height 6.0 m)

SEE Analyses (Embankment Downstream Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B3.1**

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File Name: Emb 6.0m_SEE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 18.8 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Minimum Strength: 0
Piezometric Line: 1

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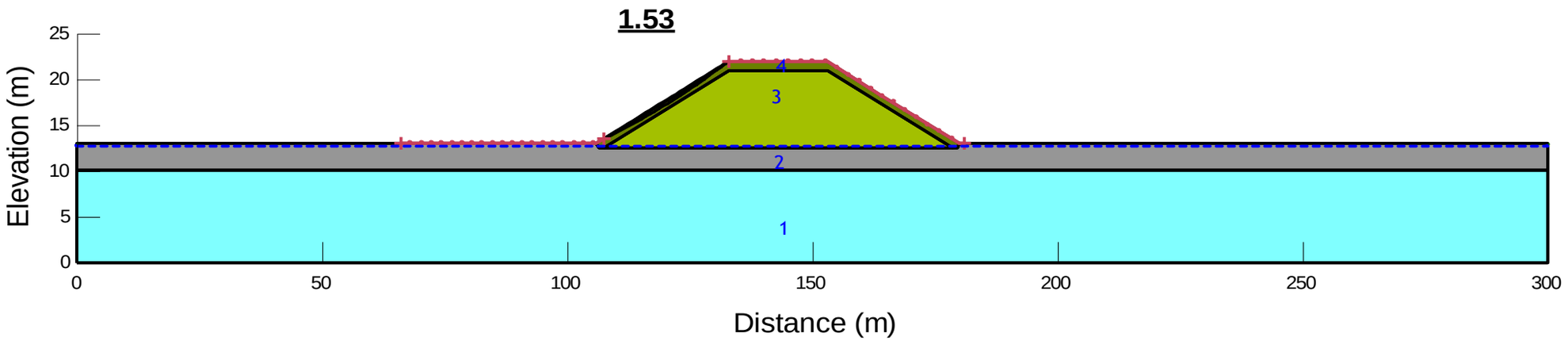
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Interior Embankment (Height 6.0 m)

SEE Analyses (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B3.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

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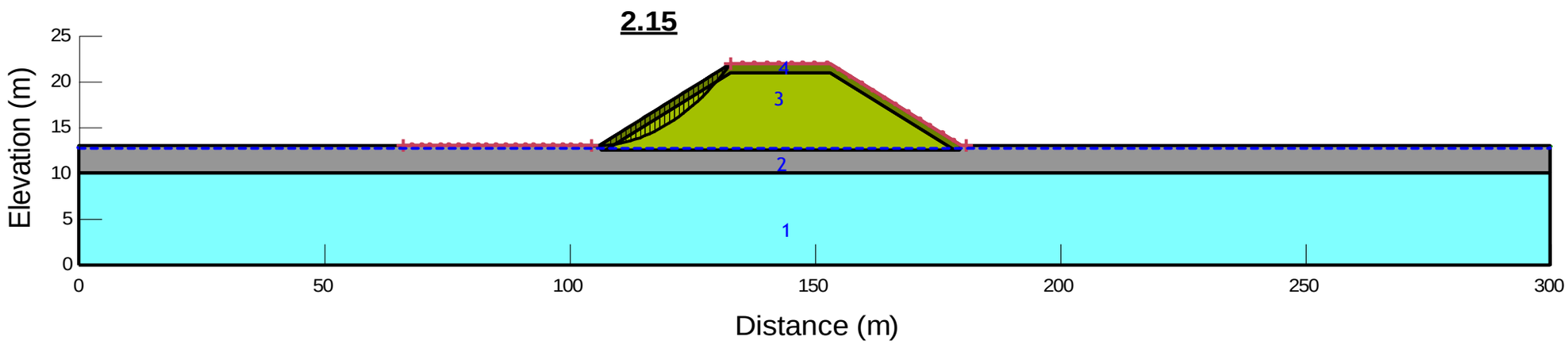
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

Static Analyses - End of Construction of Ramp (Local Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B4.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

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MANSOURAH MASSARAH GOLD MINE PROJECT

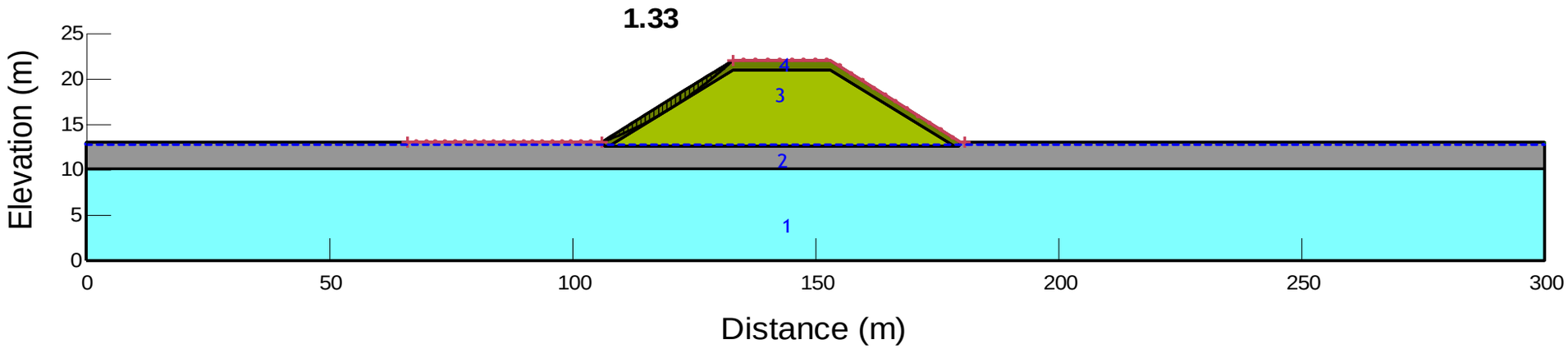
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

Static Analyses - End of Construction of Ramp (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B4.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_OBE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

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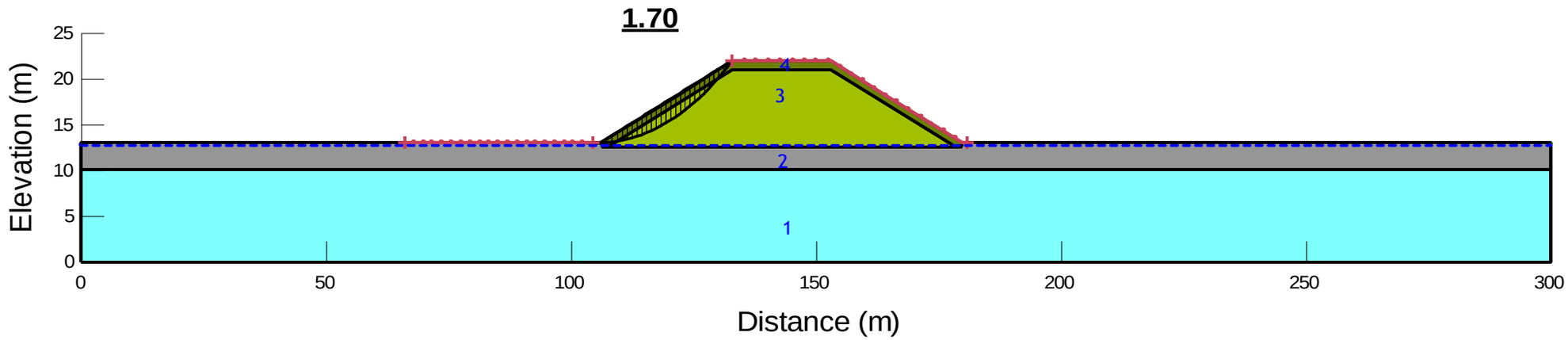
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0. m)

OBE Analyses (Local Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B5.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_OBE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

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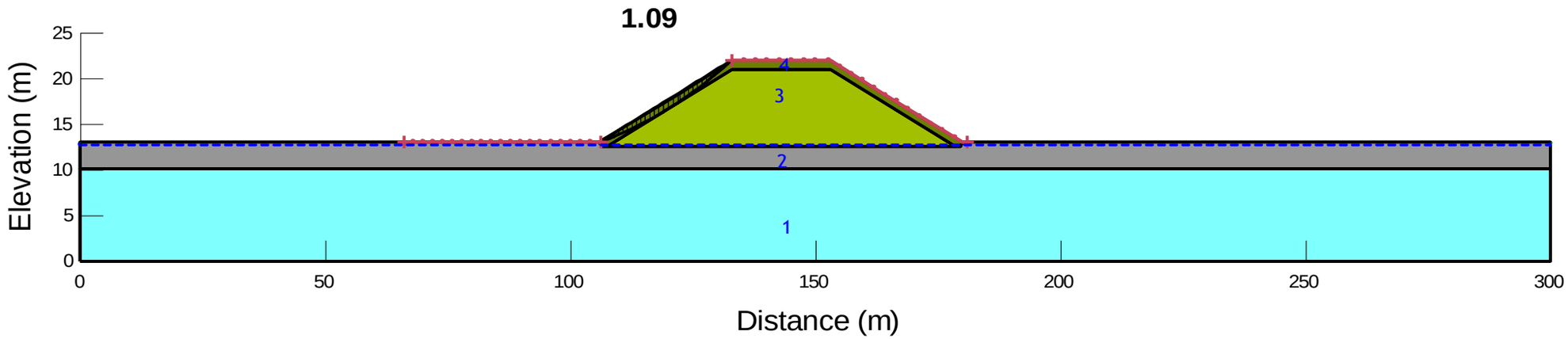
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0. m)

OBE Analyses (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B5.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_SEE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

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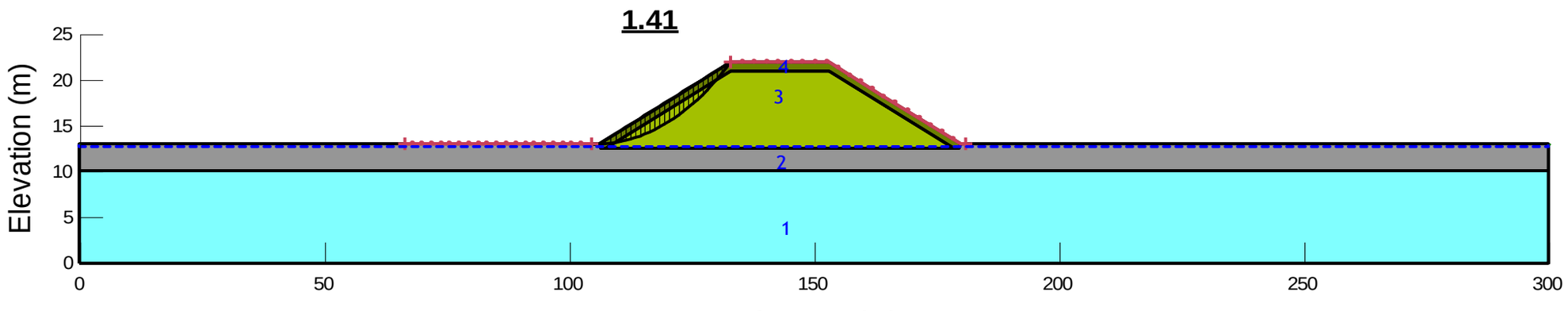
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

SEE Analyses (Local Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B6.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_SEE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

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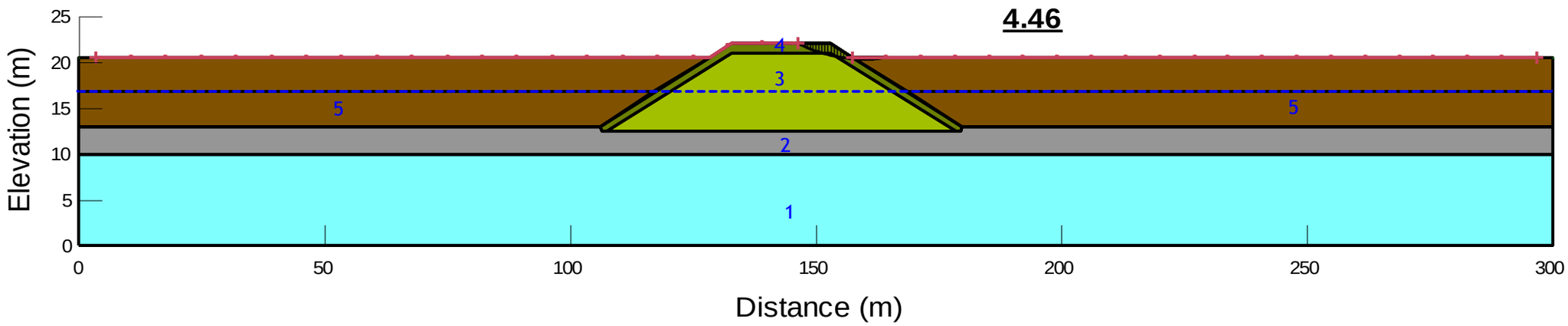
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

SEE Analyses (Global Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B6.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_Tailings_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[5] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Piezometric Line: 1

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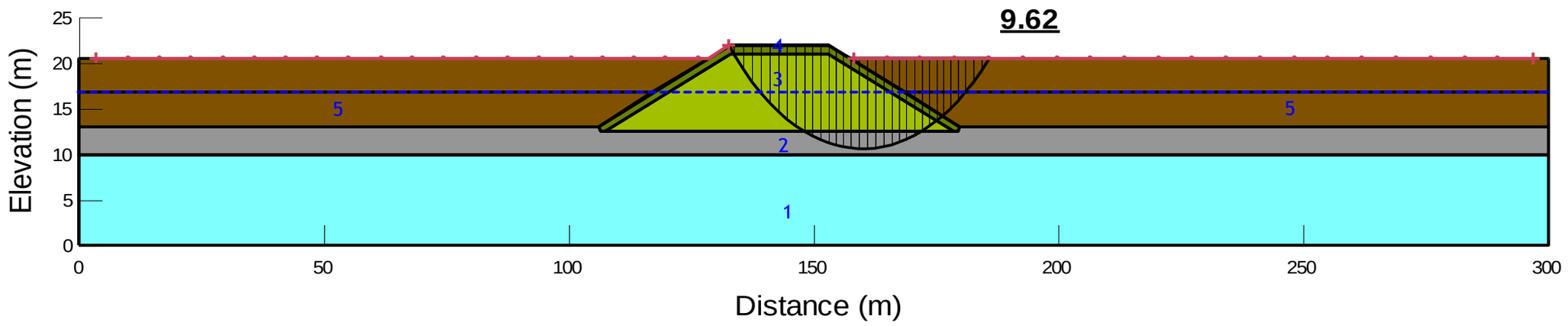
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

Static Analyses - Ramp with Tailings (Local Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B7.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_Tailings_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[5] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Piezometric Line: 1

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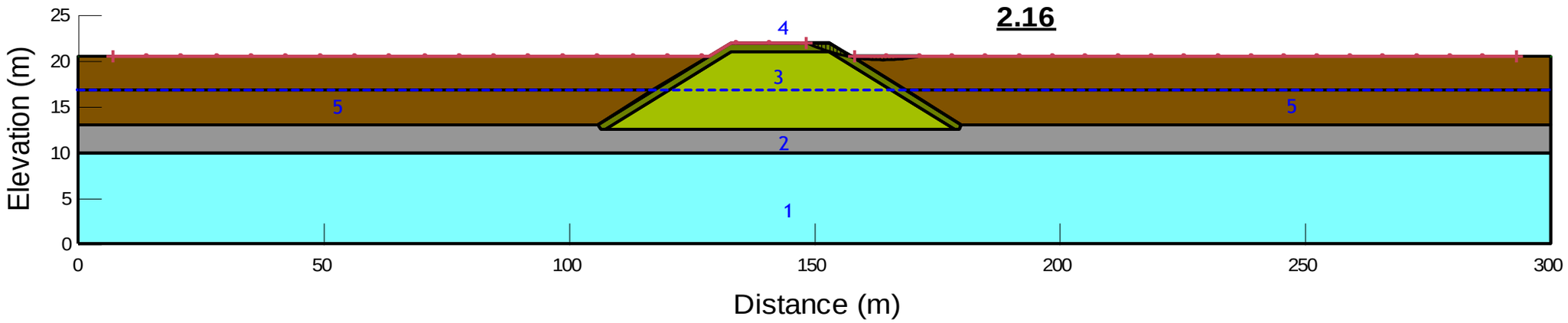
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

Static Analyses - Ramp with Tailings (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B7.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_Tailings_OBE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[5] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Piezometric Line: 1

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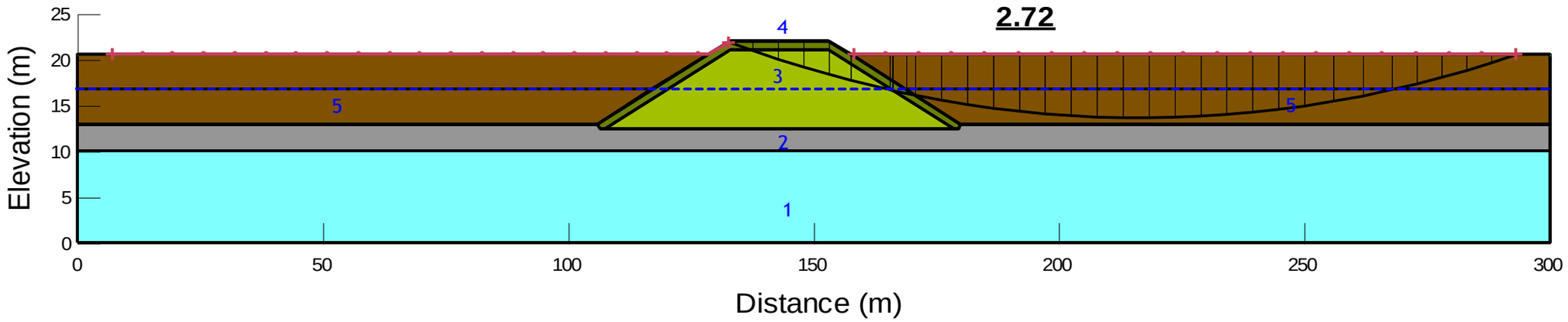
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

OBE Analyses - Ramp with Tailings (Local Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B8.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_Tailings_OBE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[5] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Piezometric Line: 1

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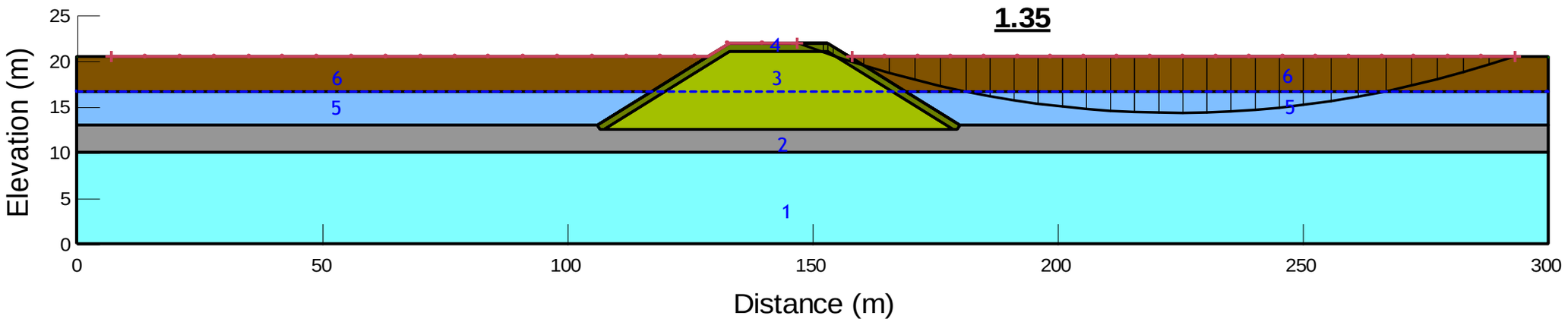
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

OBE Analyses - Ramp with Tailings (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B8.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_Tailings_SEE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.5 °
Piezometric Line: 1

[5] Name: Liquefied Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Piezometric Line: 1

[6] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.16
Minimum Strength: 0
Piezometric Line: 1

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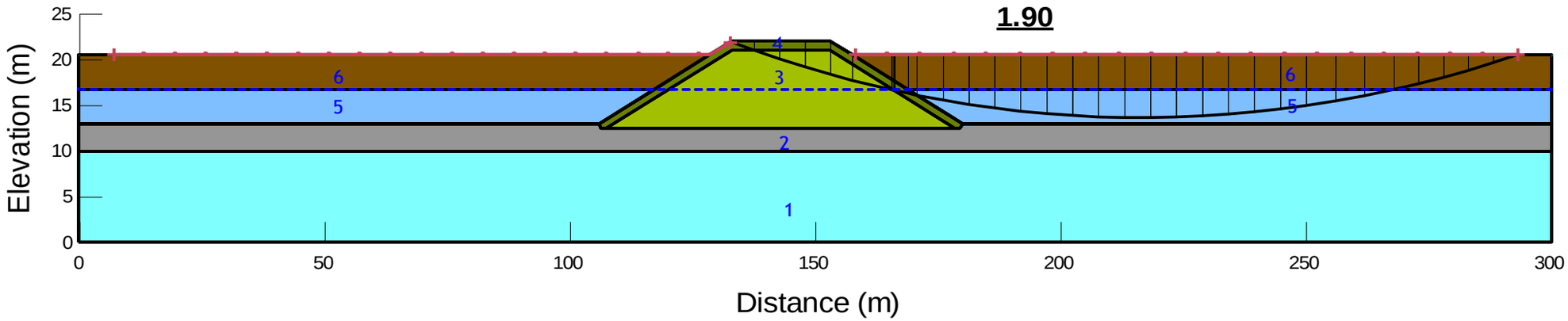
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

SEE Analyses - Ramp with Tailings (Local Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B9.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Ramp 9m_Tailings_SEE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.5 °
Piezometric Line: 1

[5] Name: Liquefied Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Piezometric Line: 1

[6] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.16
Minimum Strength: 0
Piezometric Line: 1

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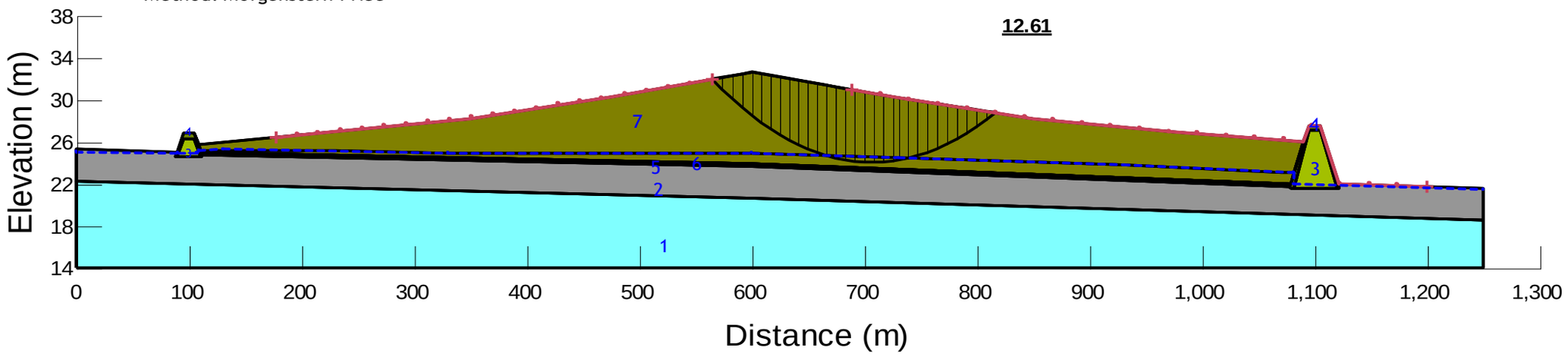
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 9.0 m)

SEE Analyses - Ramp with Tailings (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B9.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Embs Tailings_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[5] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[6] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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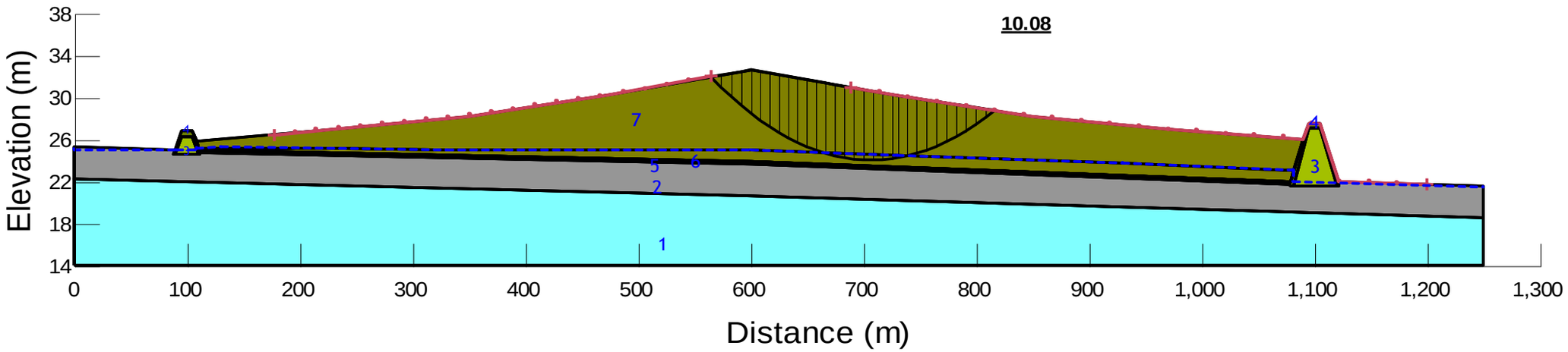
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses of TSF StartUp Embankment with Height 6.0 m and HOB 9.0 m

Static Analyses - Circular Failure (Global)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B10**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\
File Name: Embs Tailings_OBE.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[5] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 18.8 °
Piezometric Line: 1

[6] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.16
Minimum Strength: 0
Piezometric Line: 1

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DESIGN OF TAILING STORAGE FACILITY

Stability Analyses of TSF StartUp Embankment with Height 6.0 m and HOB 9.0 m

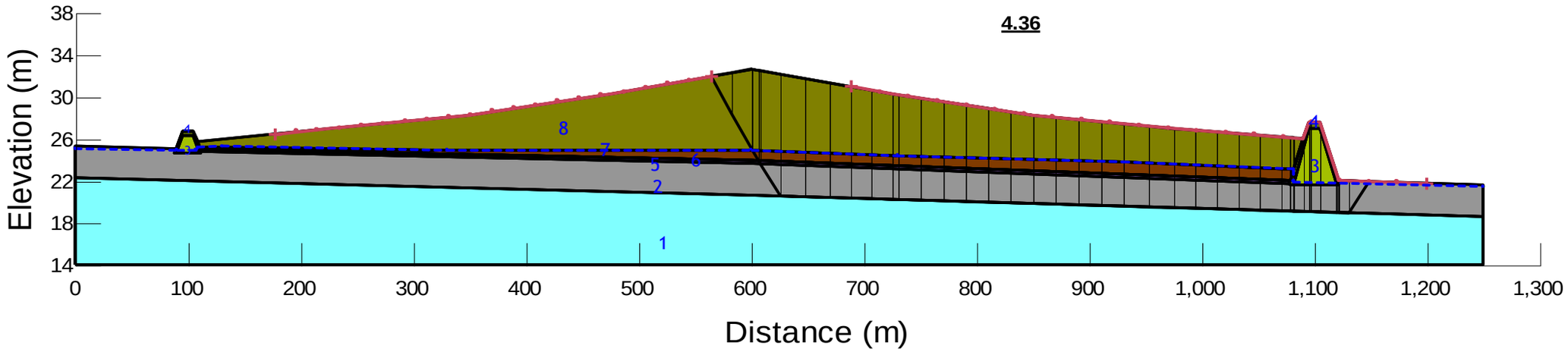
OBE Analyses - Circular Failure (Global)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B11**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Start-UP 500m\

File Name: Embs Tailings_SEE.gsz

Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: S=f(overburden)
Unit Weight: 19 kN/m³
Tau/Sigma Ratio: 0.09
Minimum Strength: 0
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[5] Name: Bedding Layer (Sand)
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.09
Minimum Strength: 0
Piezometric Line: 1

[6] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[7] Name: Liquefied Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Minimum Strength: 0
Piezometric Line: 1

[8] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.16
Minimum Strength: 0
Piezometric Line: 1

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DESIGN OF TAILING STORAGE FACILITY

Stability Analyses of TSF StartUp Embankment with Height 6.0 m and HOB 9.0 m

SEE Analyses - Circular Failure (Global)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B12**

ATTACHMENT 1.2 RADIUS OF TSF EMBANKMENT 900 M

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Job No.:	117168.03			
Calcs:	Bhagaban AC	Date:	Sep-20	
Checked:	KDS	Date:	Sep-20	

Project: Mansourah Massarrah Gold Project

Subject: Slope Stability Analyses for Intermediate TSF Embankment (Radius of Intermediate TSF Embankment = 900 m)

Software: Slope/W (GeoStudio)

Slope Stability Analyses of Intermediate TSF Facility

Height of Intermediate Embankment = 7 m

(Material parameters are assumed conservatively based on the ATCW database for the similar materials)

Layers	Unit Weight	Static / Post OBE		Post SEE	
	γ (kN/m^3)	ϕ (Deg)	s_u/σ'_v	ϕ (Deg)	s_u/σ'_v
1 Foundation (Gravelly Sand)	19	34	-	28.3	0.09
2 Bedding Layer (Sand)	16	23	-	18.8	0.09
3 Embankment (Rock)	20	35	-	29.3	-
4 Embankment (Sand)	18	27	-	22.2	-
5 Liner	10	10	-	8	-
6 Tailings	16	-	0.2		0.08

Stability Analyses Criteria:

Analyses Scenario	Factor of Safety Requirement (ANCOLD)
1 Static (Long-term - normal operating conditions)	FoS ≥ 1.5
2 OBE (Serviceability)	FoS ≥ 1.3
3 SEE (Safety)	FoS ≥ 1.2

Earthquake Acceleration (Ref: Seismic Hazard Assessment - MD-2160000-EG-MG-RPT-0007)

1 OBE Earthquake	PGA = 0.078 g
2 SEE Earthquake	PGA = 0.13 g

Result Output:

A: Embankment Stability Analyses (Ht of Embankment = 7.0 m)

Analyses Scenario	Downstream Failure FoS	Figure	Global FoS	Figure
1 Static Case (End of construction)	2.29	B13.1	3.81	B13.2
2 OBE Case	1.79	B14.1	2.67	B14.2
3 SEE Case	1.56	B15.1	2.25	B15.2

B: Ramp Stability Analyses (Ht of Ramp = 15.5 m)

Analyses Scenario	FoS	Figure
1 Static Case (End of construction)	1.44	B16

Note:

(1) Slope stability analyses result considering shallow (local) failure is not included here as it would have not significant influence in the overall stability as well as intrigity of the TSF embankment.

(2) Potential global failure as well as failure towards the downstream region of the TSF embankment is considered in the current stability analyses.

(3) Friction angle for the sand bedding layer is the assumed interface friction angle as this layer is underlain by the liner.

(4) Liner material has usually low friction angle thus 10 degree was assumed.

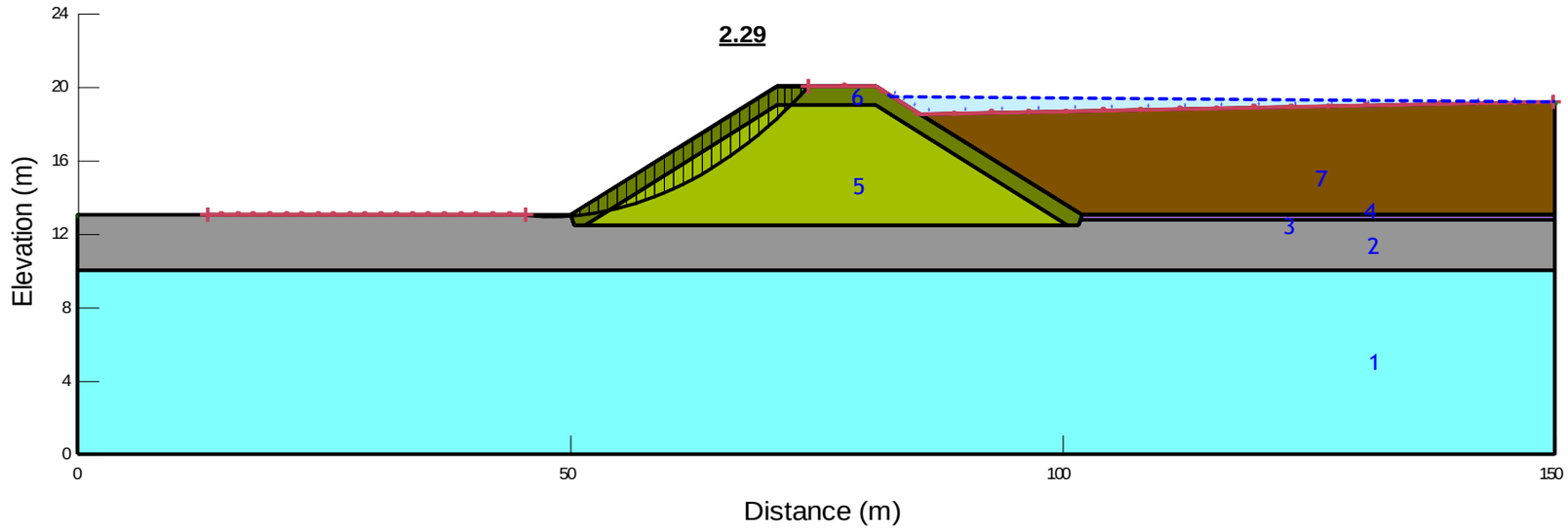
(5) For rockfill and embankment sand, friction angles were assumed conservatively based on the ATCW experiences in the similar materials.

(6) As per USCE approach, SEE analyses for embankment and ramp include reduction of strength parameters of materials by 20%, adopting liquefied strength of tailings, and adopting half of peak ground acceleration.

(7) For the TSF including tailings inside the embankments, post seismic analyses are conducted.

End

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Intermediate 900m\
File Name: IntEmb 7.0m_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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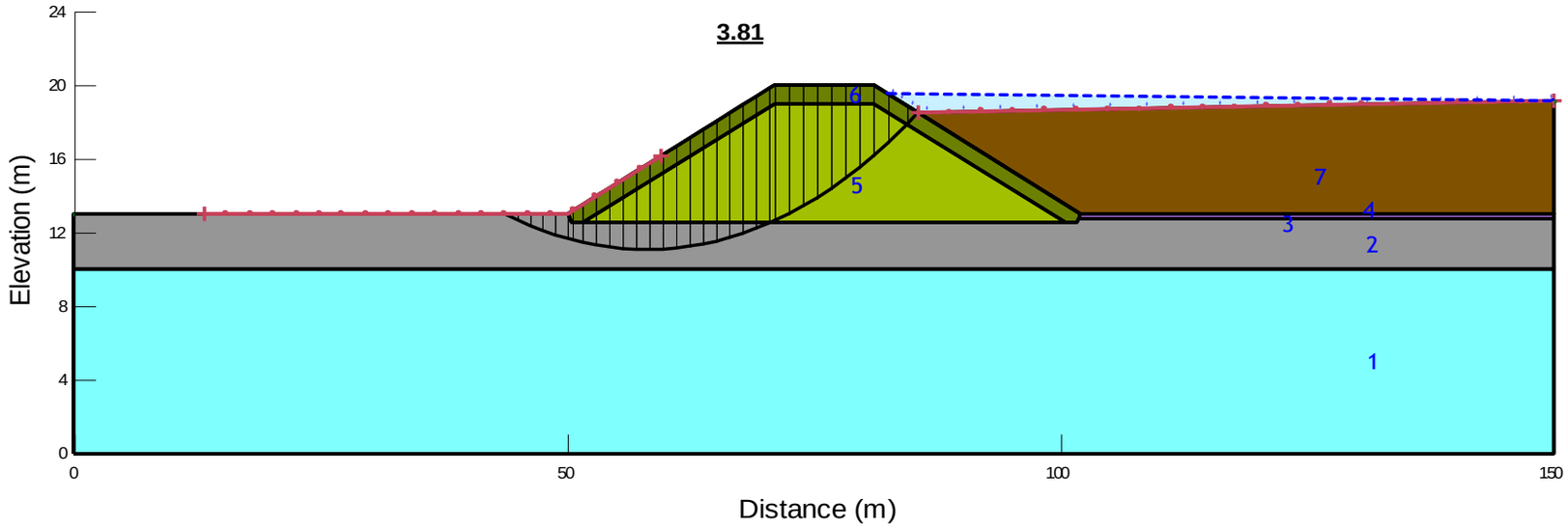
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Intermediate Embankment (Height 7.0 m)

Static Analyses (Embankment Downstream Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B13.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Intermediate 900m\
File Name: IntEmb 7.0m_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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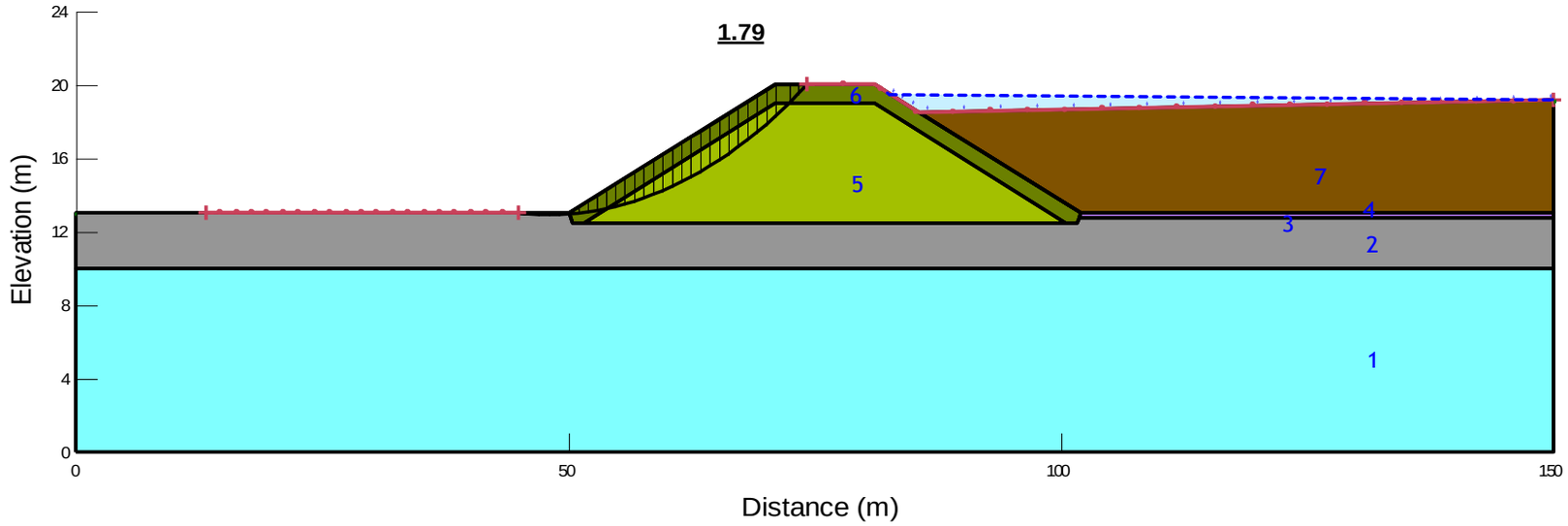
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Stability Analyses - Intermediate Embankment (Height 7.0 m)

Static Analyses (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B13.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Intermediate 900m\
File Name: IntEmb 7.0m_OBE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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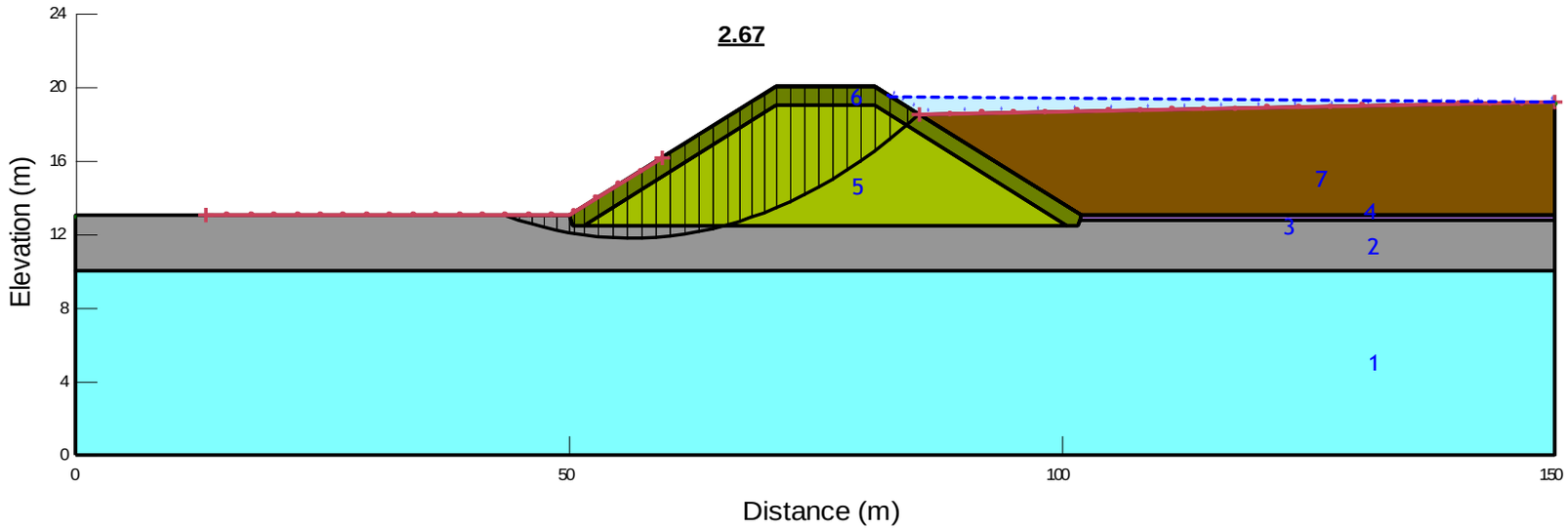
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Intermediate Embankment (Height 7.0 m)

OBE Analyses (Embankment Downstream Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B14.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Intermediate 900m\
File Name: IntEmb 7.0m_OBE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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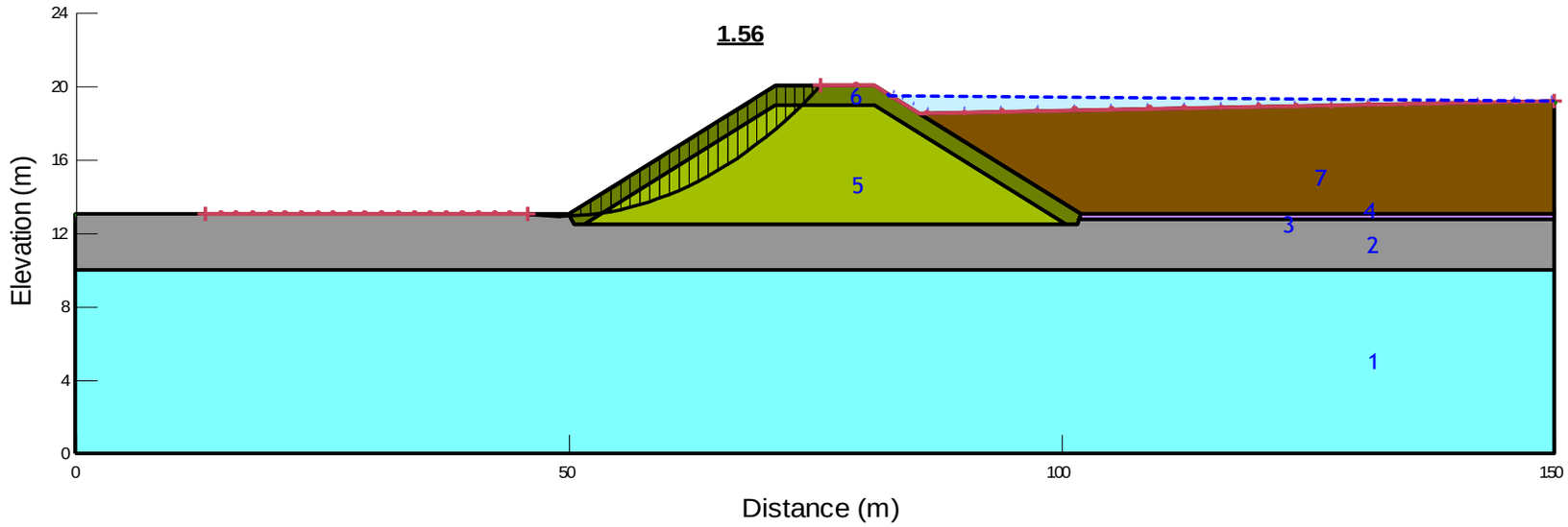
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Stability Analyses - Intermediate Embankment (Height 7.0 m)

OBE Analyses (Global Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B14.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Intermediate 900m\
File Name: IntEmb 7.0m_SEE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 18.8 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Minimum Strength: 0
Piezometric Line: 1

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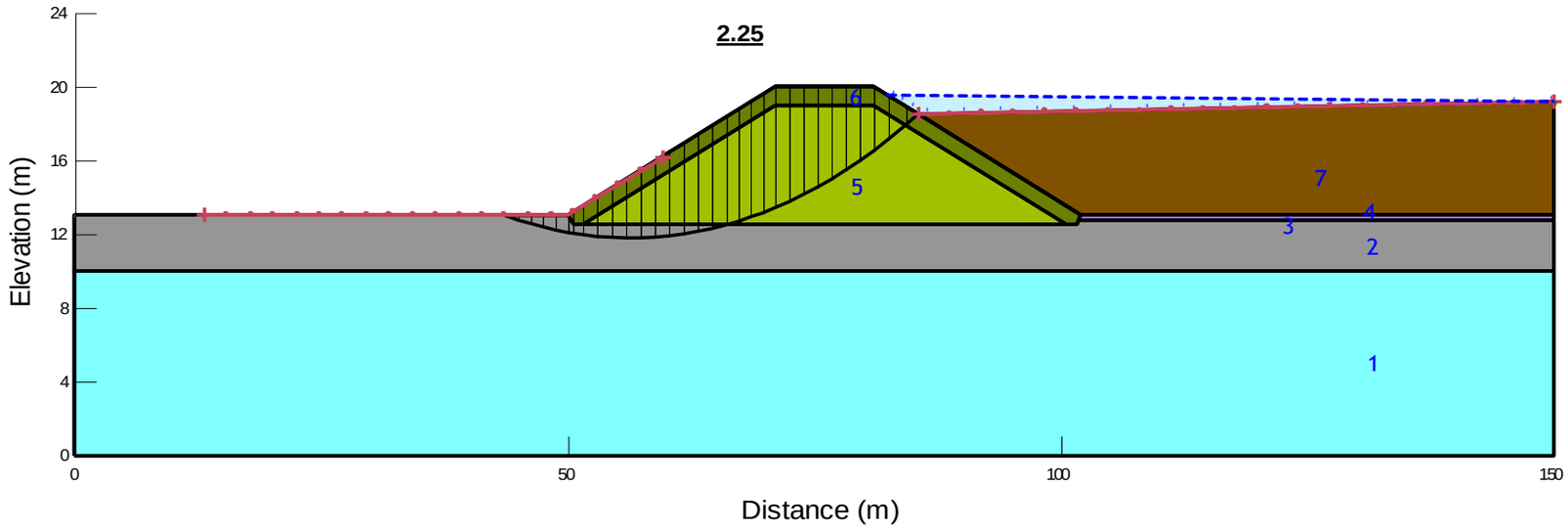
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Intermediate Embankment (Height 7.0 m)

SEE Analyses (Embankment Downstream Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B15.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Intermediate 900m\
File Name: IntEmb 7.0m_SEE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 18.8 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Minimum Strength: 0
Piezometric Line: 1

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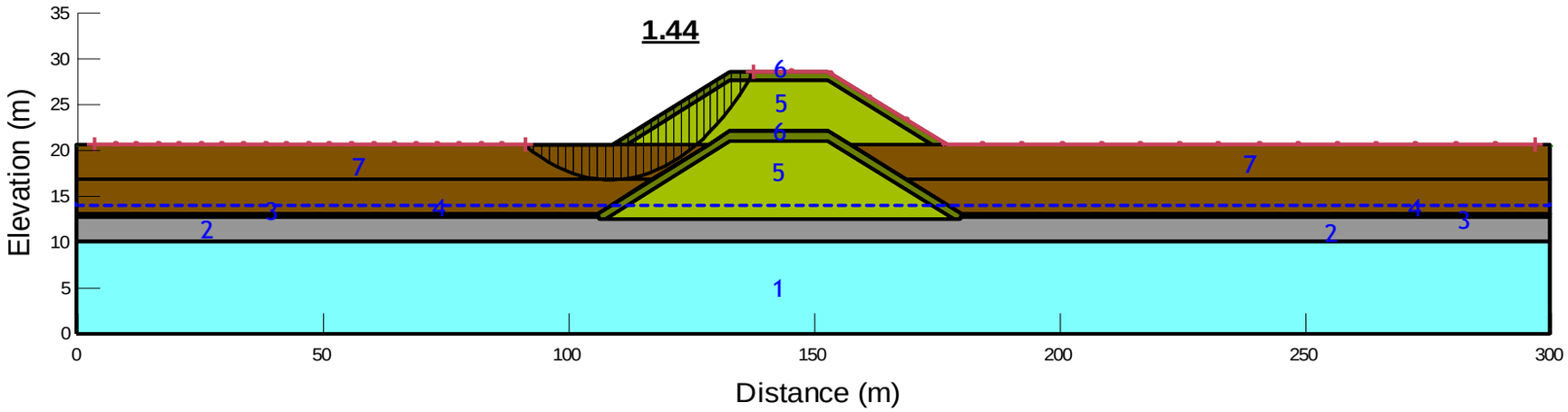
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Intermediate Embankment (Height 7.0 m)

SEE Analyses (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B15.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb Intermediate 900m\
File Name: Ramp 15.5m_Tailings_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Phi-B: 0 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Phi-B: 0 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Phi-B: 0 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Phi-B: 0 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Phi-B: 0 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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MANSOURAH MASSARAH GOLD MINE PROJECT

DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 15.5 m)

Static Analyses - End of Construction of Ramp

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B16**

ATTACHMENT 1.3 RADIUS OF TSF EMBANKMENT 1200 M

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Job No.:	117168.03			
Calcs:	Bhagaban AC	Date:	Sep-20	
Checked:	KDS	Date:	Sep-20	

Project: Mansourah Massarrah Gold Project

Subject: Slope Stability Analyses for Intermediate TSF Embankment (Radius of Intermediate TSF Embankment = 1200 m)

Software: Slope/W (GeoStudio)

Slope Stability Analyses of LOM TSF Facility

Height of LOM Embankment = 8 m

(Material parameters are assumed conservatively based on the ATCW database for the similar materials)

Layers	Unit Weight	Static / Post OBE		Post SEE	
	γ (kN/m^3)	ϕ (Deg)	s_u/σ'_v	ϕ (Deg)	s_u/σ'_v
1 Foundation (Gravelly Sand)	19	34	-	28.3	0.09
2 Bedding Layer (Sand)	16	23	-	18.8	0.09
3 Embankment (Rock)	20	35	-	29.3	-
4 Embankment (Sand)	18	27	-	22.2	-
5 Liner	10	10	-	8	-
6 Tailings	16	-	0.2		0.08

Stability Analyses Criteria:

Analyses Scenario	Factor of Safety Requirement (ANCOLD)
1 Static (Long-term - normal operating conditions)	FoS ≥ 1.5
2 OBE (Serviceability)	FoS ≥ 1.3
3 SEE (Safety)	FoS ≥ 1.2

Earthquake Acceleration (Ref: Seismic Hazard Assessment - MD-2160000-EG-MG-RPT-0007)

1 OBE Earthquake	PGA = 0.078 g
2 SEE Earthquake	PGA = 0.13 g

Result Output:

A: Embankment Stability Analyses (Ht of Embankment = 8.0 m)

Analyses Scenario	Downstream Failure FoS	Figure	Global FoS	Figure
1 Static Case (End of construction)	2.27	B17.1	3.55	B17.2
2 OBE Case	1.78	B18.1	2.54	B18.2
3 SEE Case	1.48	B19.1	2.13	B19.2

B: Ramp Stability Analyses (Ht of Ramp = 20m (2026) and 25m (2029))

Analyses Scenario	FoS	Figure
1 Static Case - Ht of Ramp = 20m m (End of construction - Year 2026)	1.39	B20
2 Static Case - Ht of Ramp = 20m m (End of construction - Year 2029)	1.34	B21

C: TSF Stability Analyses with Tailings inside Embankment during LOM (HOB = 23.5 m)

Analyses Scenario	FoS	Figure
1 Static Case - Tailings Beach Slope Failure	7.61	B22
2 Post-OBE Case - Tailings Beach Slope Failure	5.09	B23
3 Post-SEE Case - Tailings Beach Slope Failure	3.09	B24

(1) Slope stability analyses result considering shallow (local) failure is not included here as it would have not significant influence in the overall stability as well as intrigity of the TSF embankment.

(2) Potential global failure as well as failure towards the downstream region of the TSF embankment is considered in the current stability analyses.

(3) Friction angle for the sand bedding layer is the assumed interface friction angle as this layer is underlain by the liner.

(4) Liner material has usually low friction angle thus 10 degree was assumed.

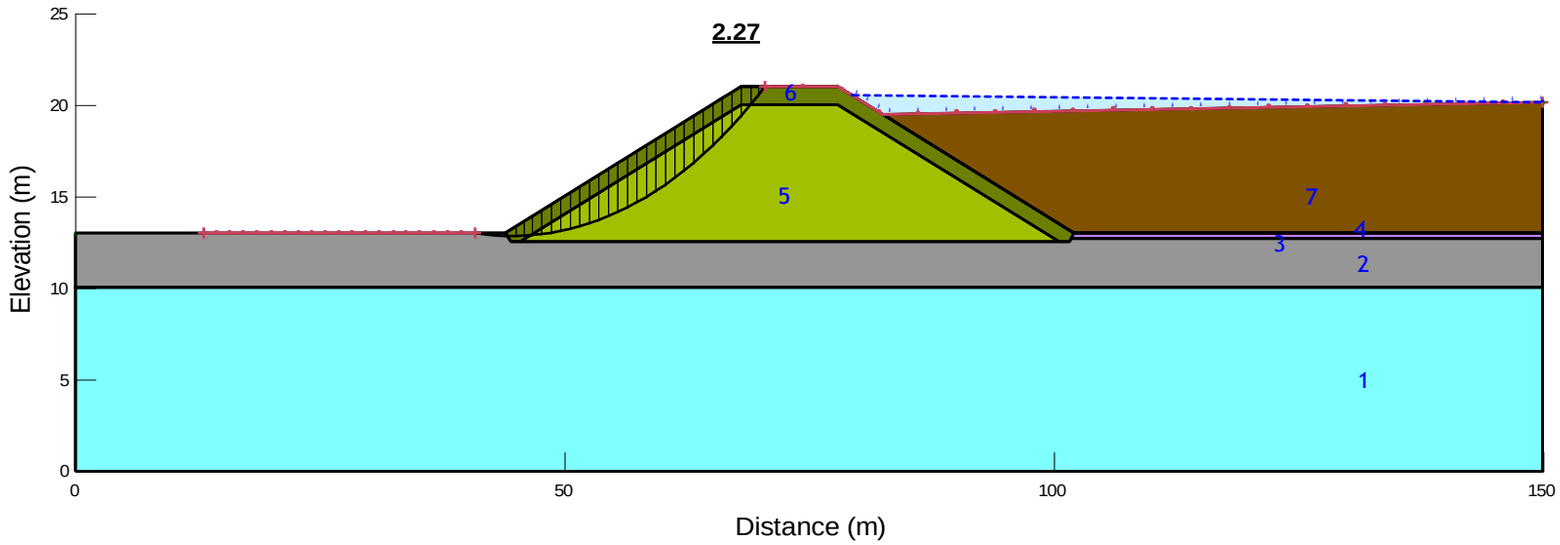
(5) For rockfill and embankment sand, friction angles were assumed conservatively based on the ATCW experiences in the similar materials.

(6) As per USCE approach, SEE analyses for embankment and ramp include reduction of strength parameters of materials by 20%, adopting liquefied strength of tailings, and adopting half of peak ground acceleration.

(7) For the TSF including tailings inside the embankments, post seismic analyses are conducted.

End

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\
File Name: LOMEmb 8m_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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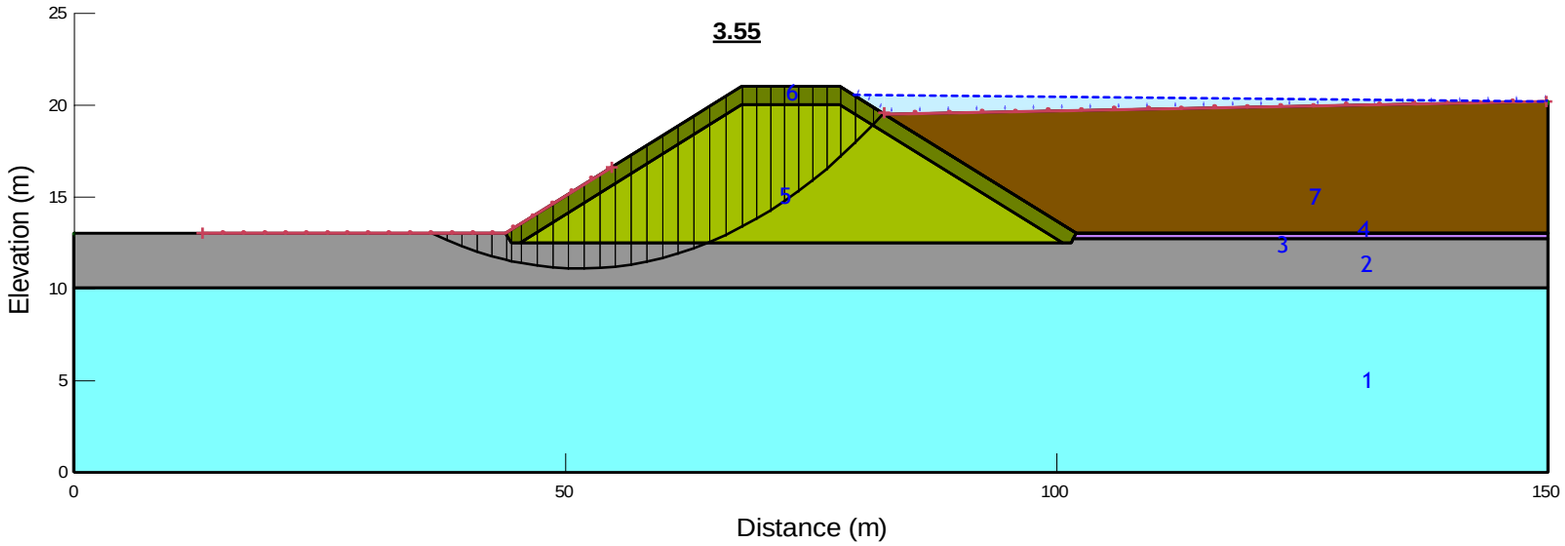
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - LOM Embankment (Height 8.0 m)

Static Analyses (Embankment Downstream Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B17.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\
File Name: LOMEmb 8m_Static.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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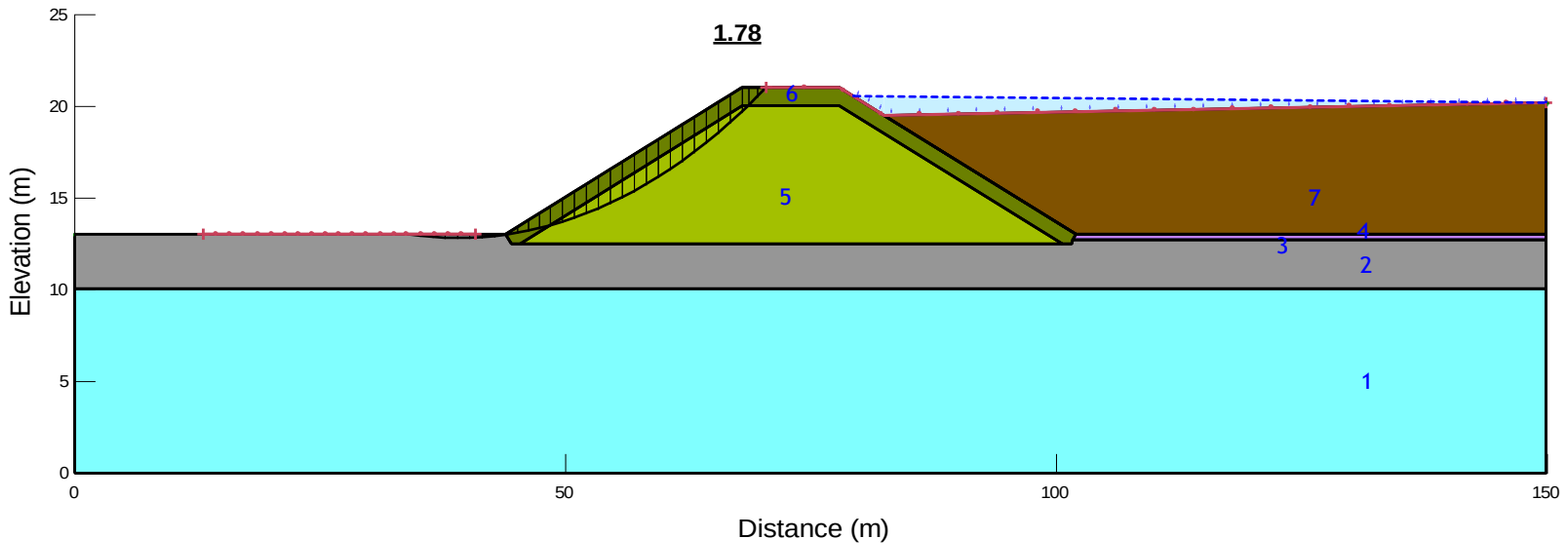
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - LOM Embankment (Height 8.0 m)

Static Analyses (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B17.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\
File Name: LOMEmb 8m_OBE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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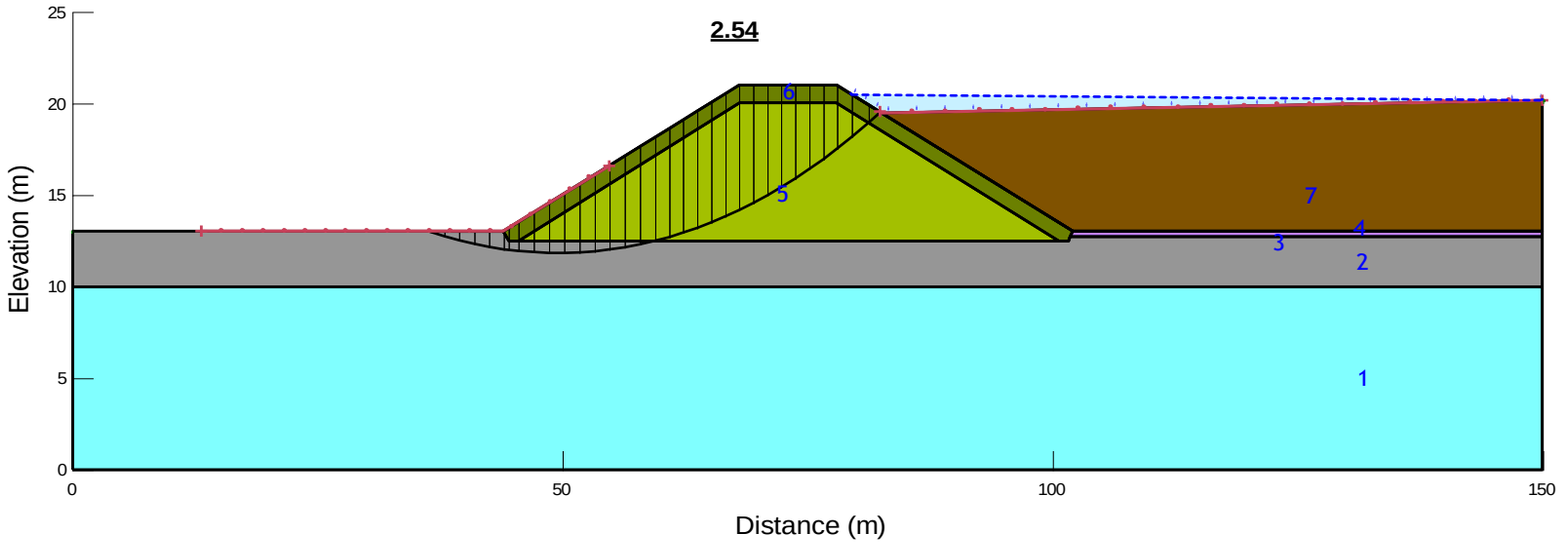
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - LOM Embankment (Height 8.0 m)

OBE Analyses (Embankment Downstream Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B18.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\
File Name: LOMEmb 8m_OBE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.078



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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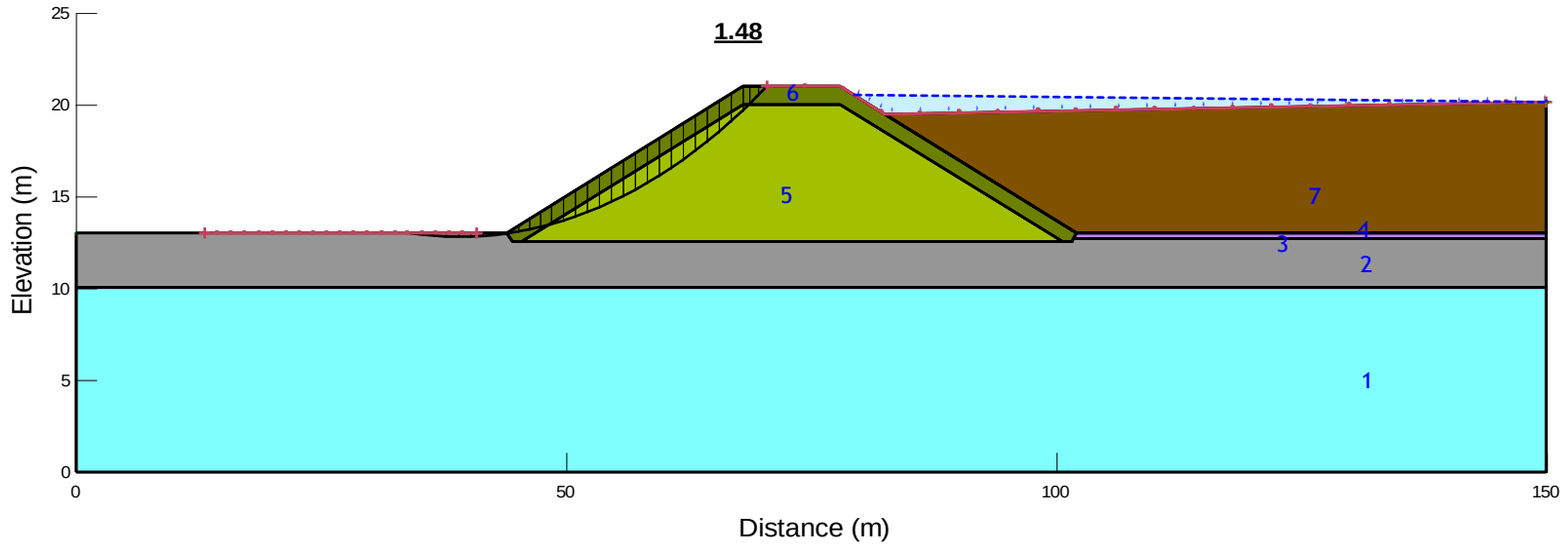
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - LOM Embankment (Height 8.0 m)

OBE Analyses (Global Failure)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B18.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\
File Name: LOMEmb 8m_SEE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 18.8 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Minimum Strength: 0
Piezometric Line: 1

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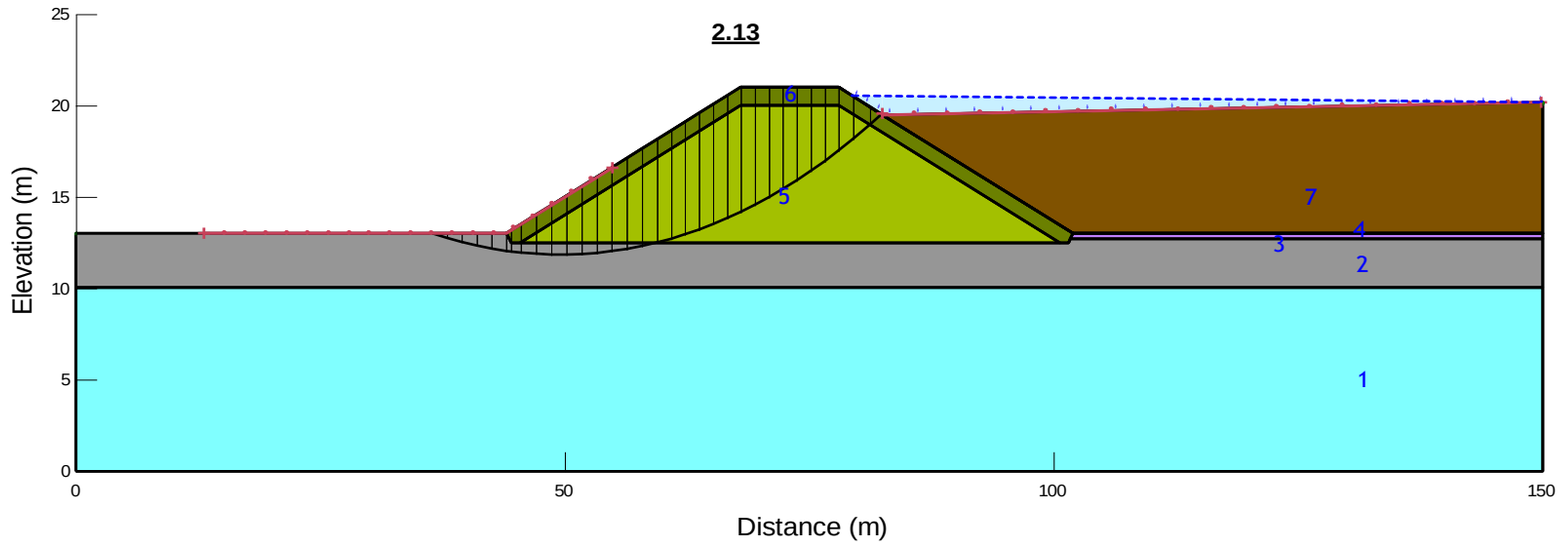
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - LOM Embankment (Height 8.0 m)

SEE Analyses (Embankment Downstream Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B19.1**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\
File Name: LOMEmb 8m_SEE.gsz
Method: Morgenstern-Price
Horz Seismic Coef.: 0.065



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 18.8 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Minimum Strength: 0
Piezometric Line: 1

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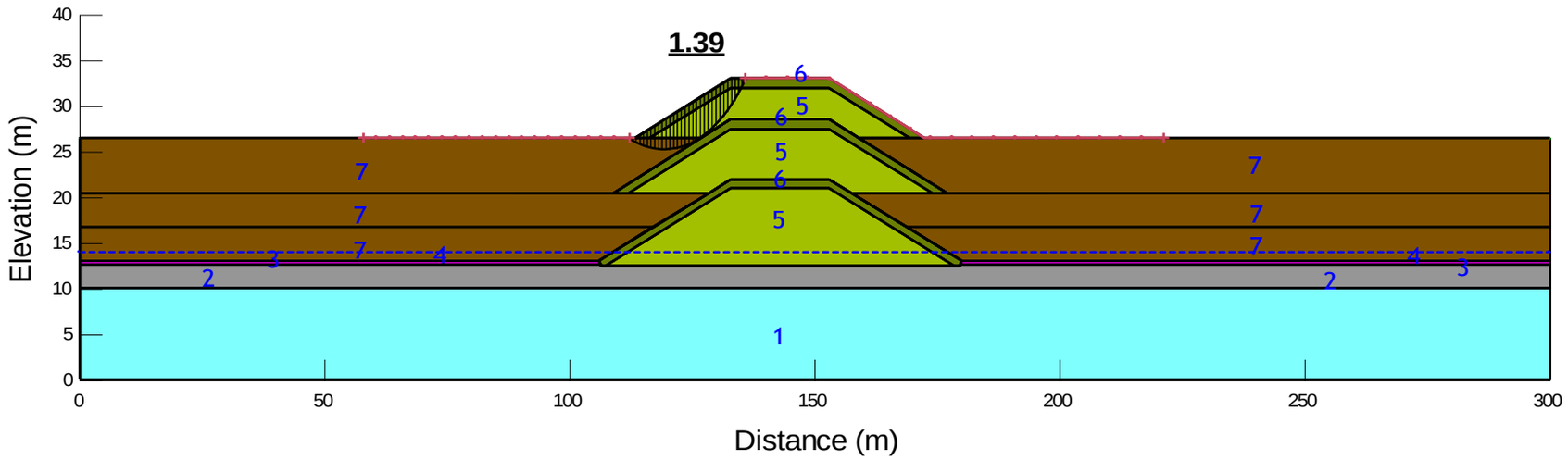
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - LOM Embankment (Height 8.0 m)

SEE Analyses (Global Failure)

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B19.2**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\
File Name: Ramp 20m_Tailings_Static_2026.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Phi-B: 0 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Phi-B: 0 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Phi-B: 0 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Phi-B: 0 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Phi-B: 0 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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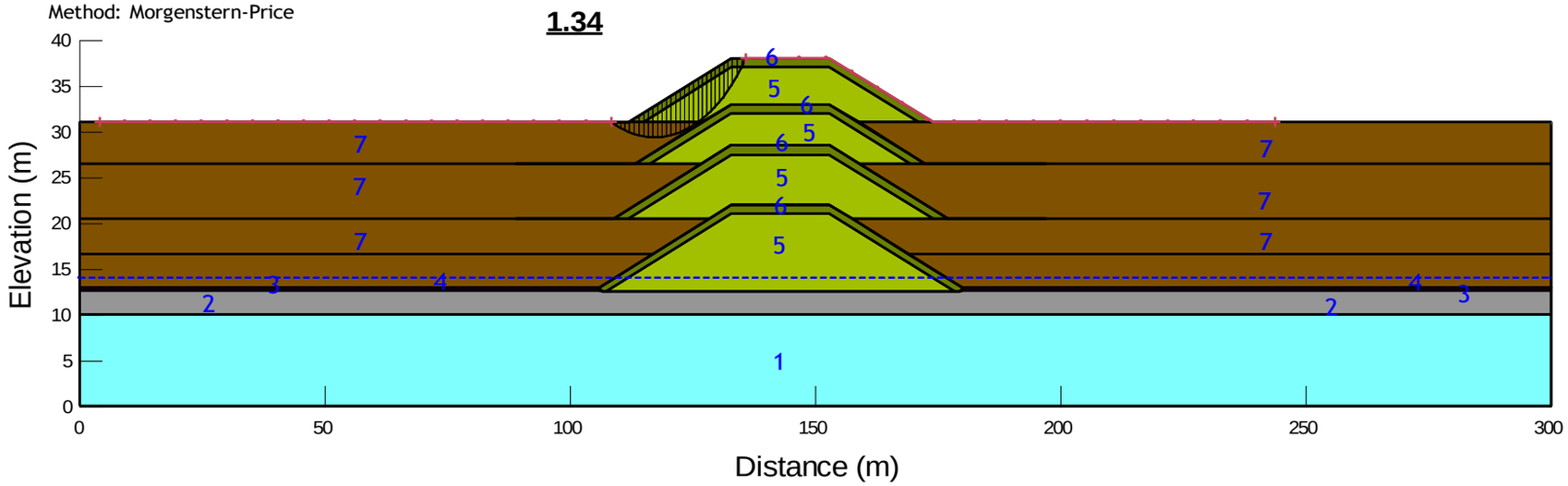
DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 20 m)

Static Analyses - End of Construction of Ramp (2026)

Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B20**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\
File Name: Ramp 25m_Tailings_Static_2029.gsz
Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Phi-B: 0 °
Piezometric Line: 1

[3] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Phi-B: 0 °
Piezometric Line: 1

[4] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Phi-B: 0 °
Piezometric Line: 1

[5] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Phi-B: 0 °
Piezometric Line: 1

[6] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Phi-B: 0 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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DESIGN OF TAILING STORAGE FACILITY

Stability Analyses - Ramp (Height 25 m)

Static Analyses - End of Construction of Ramp (2029)

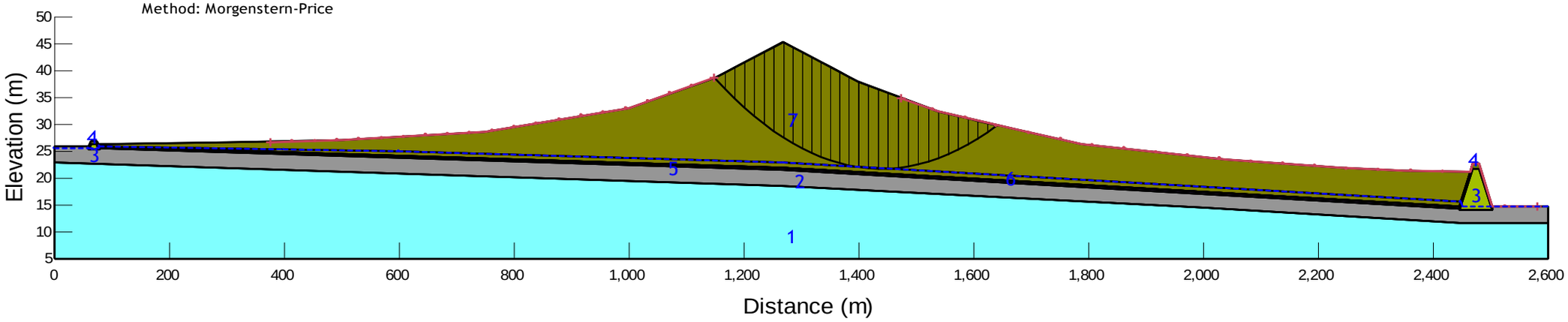
Date: 10/09/2020	**Job No:** 117168.03	**FIGURE B21**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\Liq Beach\

File Name: Embs Liq LOM_Static.gsz

7.61

Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 34 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 35 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 27 °
Piezometric Line: 1

[5] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 23 °
Piezometric Line: 1

[6] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 10 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.2
Minimum Strength: 0
Piezometric Line: 1

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DESIGN OF TAILING STORAGE FACILITY

Stability Analyses of LOM Embankment with Height 8 m and HOB 23.5 m

Static Analyses - Tailings Beach Slope Failure

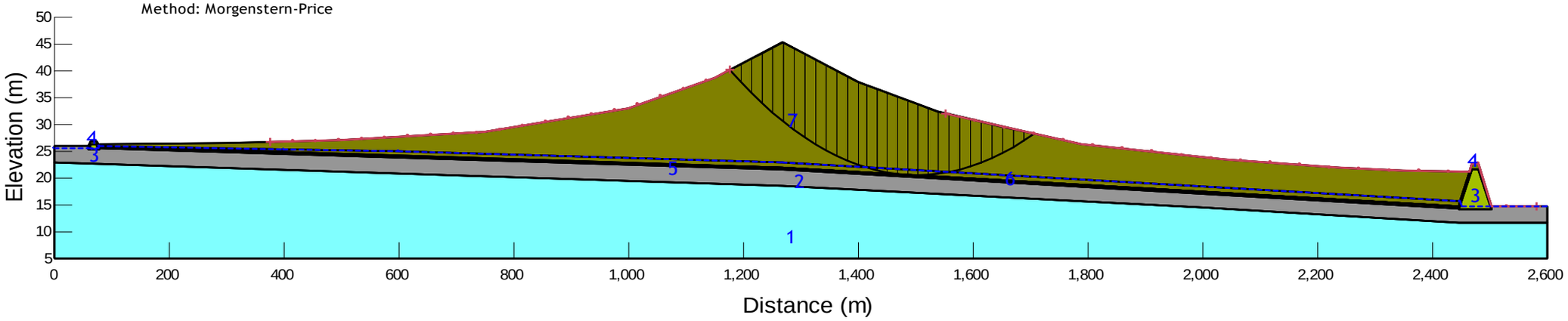
Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B22**

Directory: K:\Projects\117\117168 Mansourah-Massarah Gold Project\03- Detailed Design\Data and Calcs\SlopeW\BA\Emb LOM 1200m\Liq Beach\

File Name: Embs Liq LOM_OBE.gsz

5.09

Method: Morgenstern-Price



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: Mohr-Coulomb
Unit Weight: 19 kN/m³
Cohesion': 0 kPa
Phi': 28.3 °
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[5] Name: Bedding Layer (Sand)
Model: Mohr-Coulomb
Unit Weight: 16 kN/m³
Cohesion': 0 kPa
Phi': 18 °
Piezometric Line: 1

[6] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[7] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.16
Minimum Strength: 0
Piezometric Line: 1

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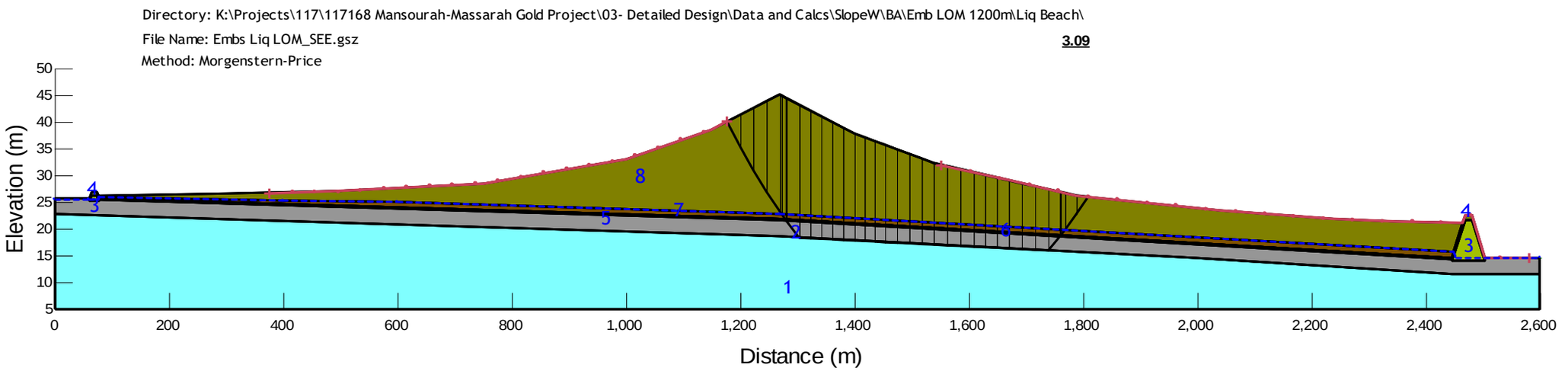
MANSOURAH MASSARAH GOLD MINE PROJECT

DESIGN OF TAILING STORAGE FACILITY

Stability Analyses of LOM Embankment with Height 8 m and HOB 23.5 m

OBE Analyses - Tailings Beach Slope Failure

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B23**



[1] Name: Bed Rock
Model: Bedrock (Impenetrable)
Piezometric Line: 1

[2] Name: Foundation (Gravelly Sand)
Model: S=f(overburden)
Unit Weight: 19 kN/m³
Tau/Sigma Ratio: 0.09
Minimum Strength: 0
Piezometric Line: 1

[3] Name: Embankment (Rock)
Model: Mohr-Coulomb
Unit Weight: 20 kN/m³
Cohesion': 0 kPa
Phi': 29.3 °
Piezometric Line: 1

[4] Name: Embankment (Sand)
Model: Mohr-Coulomb
Unit Weight: 18 kN/m³
Cohesion': 0 kPa
Phi': 22.2 °
Piezometric Line: 1

[5] Name: Bedding Layer (Sand)
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.09
Minimum Strength: 0
Piezometric Line: 1

[6] Name: Liner
Model: Mohr-Coulomb
Unit Weight: 10 kN/m³
Cohesion': 0 kPa
Phi': 8 °
Piezometric Line: 1

[7] Name: Liquefied Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.08
Minimum Strength: 0
Piezometric Line: 1

[8] Name: Tailings
Model: S=f(overburden)
Unit Weight: 16 kN/m³
Tau/Sigma Ratio: 0.16
Minimum Strength: 0
Piezometric Line: 1

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MANSOURAH MASSARAH GOLD MINE PROJECT

DESIGN OF TAILING STORAGE FACILITY

Stability Analyses of LOM Embankment with Height 8 m and HOB 23.5 m

SEE Analyses - Tailings Beach Slope Failure

Date:	10/09/2020	**Job No:**	117168.03	**FIGURE B24**